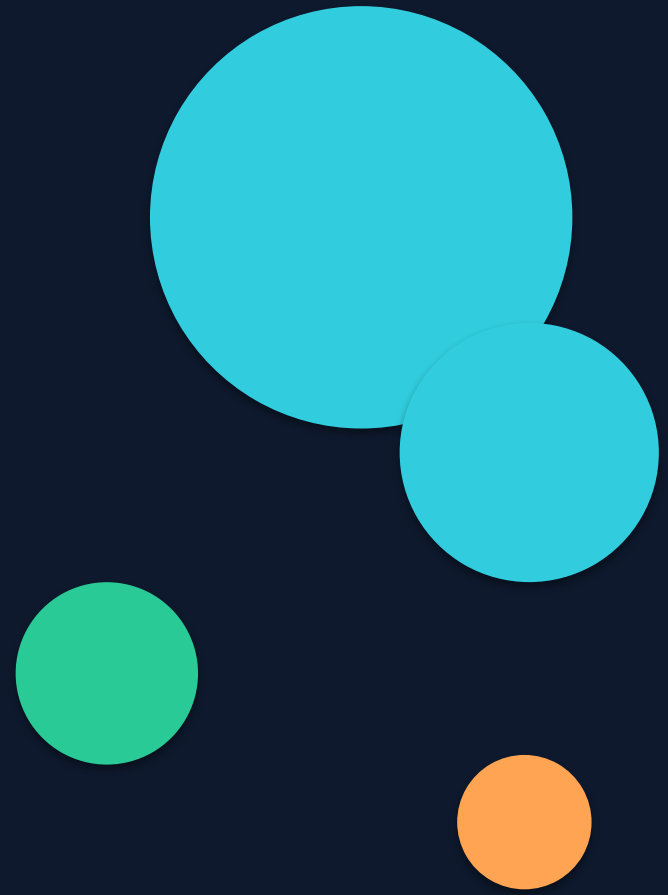
Data Collection & Data Preprocessing

Detailed compact notes with concepts, methods, examples and practical reminders.



6 topic slides + cover = 7 slides

Data Sources and Collection Methods

Key concepts • practical context • quick revision

- Primary data is collected directly through surveys, interviews, experiments or sensors.
- Secondary data comes from existing databases, reports, public datasets or organizational records.
- Common formats: CSV, Excel, JSON, relational tables, logs, images and time-series sensor data.
- APIs provide structured access to services; databases support querying large, organized collections.
- Check relevance, accuracy, coverage, freshness, cost, access permissions and collection bias.
- Document data owner, collection date, units, definitions, sampling method and known limitations.

EXAMPLE / REMEMBER

Before collecting: define the question, required fields, population and permission to use the data.

Understanding Data Structure

Key concepts • practical context • quick revision

- A row should represent a clearly defined entity or event; columns should have consistent meanings.
- Create a data dictionary describing each field, type, unit, allowed range and missing-value convention.
- Unique identifiers help detect repeated entities and link datasets.
- Distinguish categorical, numeric, ordinal, date/time, text and geographic fields.
- Check whether records cover the right time period and whether important groups are underrepresented.
- Use validation rules at collection time to prevent avoidable quality problems.

EXAMPLE / REMEMBER

Example data dictionary: attendance_pct = numeric, unit %, valid range 0–100.

Missing Values and Duplicates

Key concepts • practical context • quick revision

- Find missing values with isna()/isnull(); count them per column and row.
- Investigate why values are missing before choosing a treatment.
- Options: remove records, fill with mean/median/mode, use a domain rule or add a missingness indicator.
- Mean imputation can be distorted by outliers; median is often more robust for skewed numeric data.
- Detect exact and key-based duplicates; decide which record is authoritative before removing any.
- Document how many values/records were changed and why.

EXAMPLE / REMEMBER

```
Pandas:
print(df.isna().sum())
print(df.duplicated().sum())
df = df.drop_duplicates()
```

Outliers, Errors and Inconsistent Data

Key concepts • practical context • quick revision

- Validate impossible values such as negative age, marks above the maximum or dates in the future.
- Standardize text categories: 'NY', 'New York' and 'new york' may refer to the same category.
- Normalize units before comparing measurements, such as centimeters versus meters.
- Outliers can indicate entry errors, unusual events or valid rare cases; investigate before deleting.
- Use range checks, box plots, robust statistics and domain knowledge to flag suspicious values.
- Correct errors from trustworthy sources; do not silently replace values without a reason.

EXAMPLE / REMEMBER

Quality checks should distinguish a true unusual observation from a data-entry mistake.

Transformation and Feature Engineering

Key concepts • practical context • quick revision

- Convert columns to appropriate data types and parse dates before analysis.
- Encode categorical variables using one-hot, ordinal or other suitable encoding.
- Scale numeric features with standardization or min-max scaling when the algorithm benefits from it.
- Feature engineering derives useful inputs, such as age from birth date or total spend from item prices.
- Combine tables with joins; validate keys, unmatched rows and row counts after merging.
- Text processing may include normalization, tokenization and removal of irrelevant noise.

EXAMPLE / REMEMBER

Example: order_value = quantity × unit_price; verify units and currency first.

Preprocessing Pipeline and Leakage

Key concepts • practical context • quick revision

- Split data into training, validation and test sets according to the task and time structure.
- Fit imputation, scaling and feature selection only on training data; apply learned settings to other sets.
- Data leakage occurs when training uses information unavailable at prediction time.
- For time-series prediction, do not randomly mix future observations into training if it violates chronology.
- Automated pipelines make preprocessing consistent and easier to reproduce.
- After preprocessing, re-check shapes, types, missing values, ranges and target distribution.

EXAMPLE / REMEMBER

Rule: the test set is for final evaluation —not repeated model selection.